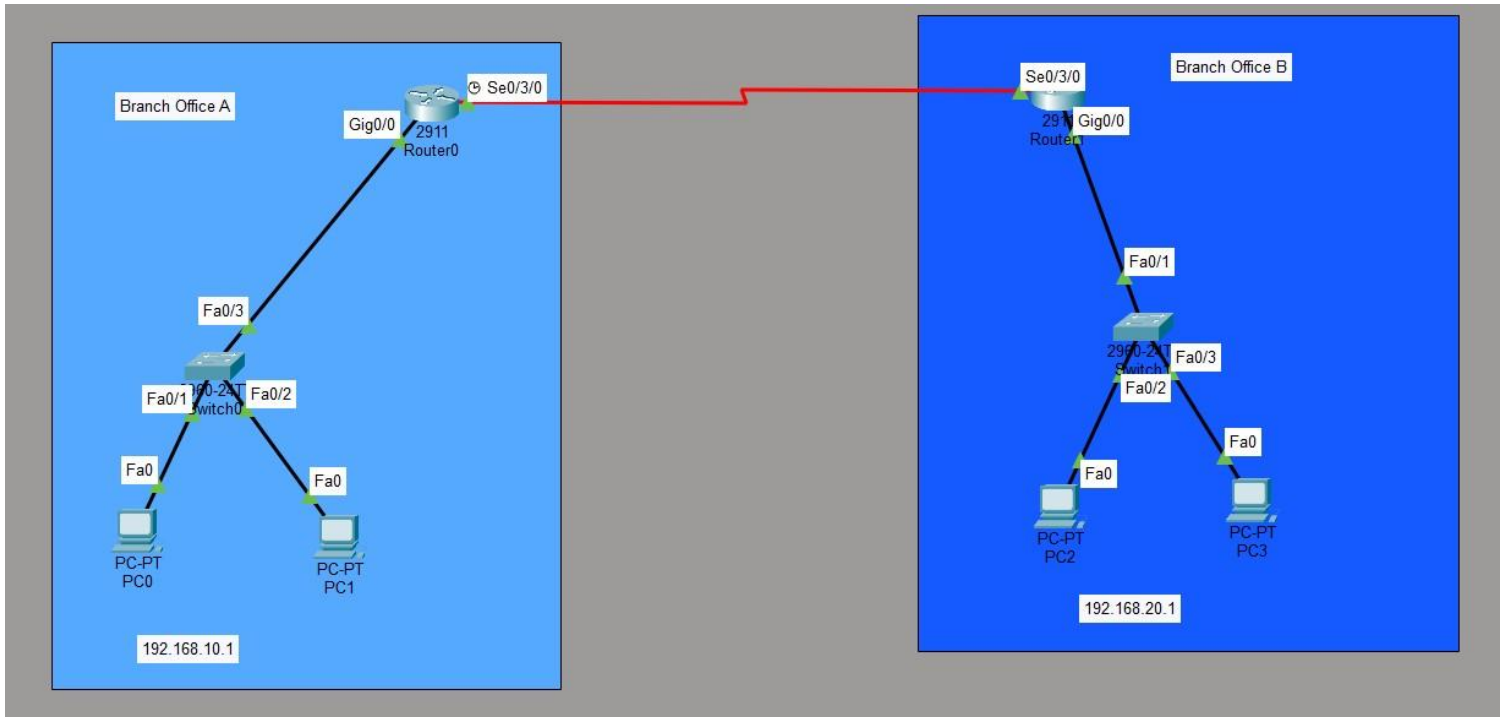


Tasks'

1. Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.



Router 0

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

Router 1

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
```

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	G0/0	192.168.10.1	255.255.255.0	—
Router0	G0/1	10.0.0.1	255.255.255.252	—
PC0	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.1
Router1	G0/0	192.168.20.1	255.255.255.0	—
Router1	G0/1	10.0.0.2	255.255.255.252	—
PC2	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1
PC3	FastEthernet0	192.168.20.11	255.255.255.0	192.168.20.1

Verification

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time=1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

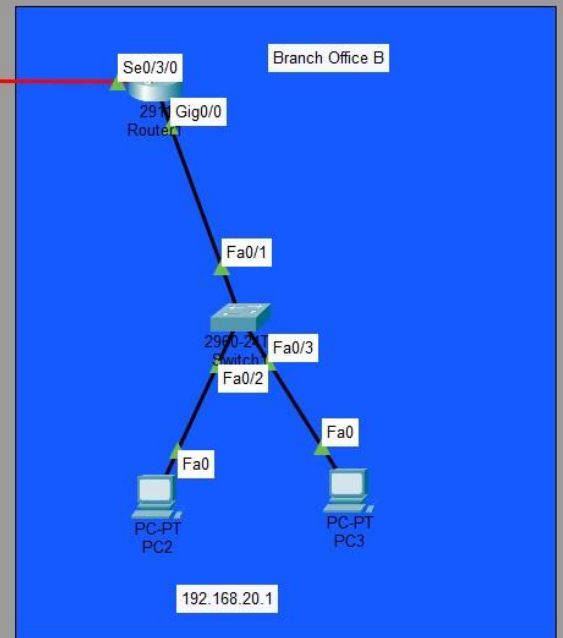
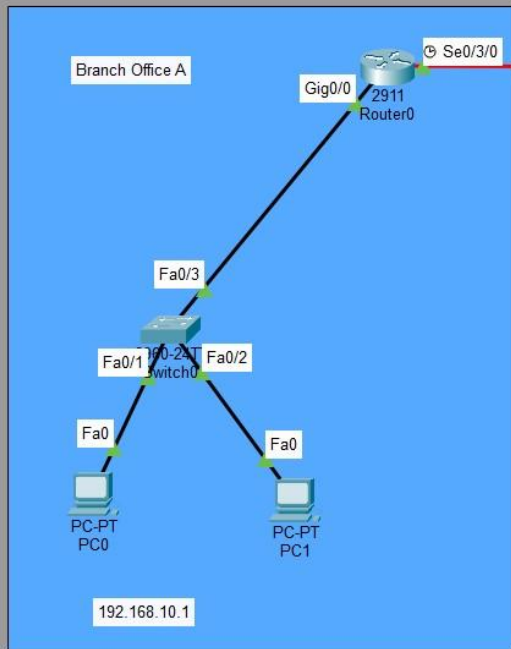
C:\>

```

The task was completed successfully. Both branch offices were configured with separate IP networks, and each branch could ping its own devices.

2. **Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.**
Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.





To configure a basic site-to-site VPN between two branch office routers using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

Create two separate branch office networks.

Configure IP addresses on routers and PCs.

Establish a GRE tunnel between Router0 and Router1.

Assign IP addresses to the tunnel interfaces.

Configure static routes for branch-to-branch communication.

Router 0

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#int tunnel 0

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

Router(config-if)#ip address 172.16.0.1 255.255.255.252
Router(config-if)#tunnel source se 0/3/0
Router(config-if)#tunnel destination 10.0.0.2
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

Router(config-if)#no shut
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.20.0 255.255.255.0 172.16.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
```

Router 1

```
Router1
Physical Config CLI Attributes

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#int tunnel 0

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

Router(config-if)#ip address 172.16.0. 255.255.255.252
      ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 172.16.0.2 255.255.255.252
Router(config-if)#tunnel source se 0/3/0
Router(config-if)#tunnel destination 10.0.0.1
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

Router(config-if)#no shut
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.10.0 255.255.255.0 172.16.0.1
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
```

Verify the GRE tunnel in router 0

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 192.168.10.1    YES manual up          up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Serial10/3/0      10.0.0.1        YES manual up          up
Serial10/3/1      unassigned      YES unset  administratively down down
Tunnel0          172.16.0.1      YES manual up          up
Vlan1            unassigned      YES unset  administratively down down
Router#
```

Verify the GRE tunnel in router 1

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 192.168.20.1    YES manual up          up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Serial10/3/0      10.0.0.2        YES manual up          up
Serial10/3/1      unassigned      YES unset  administratively down down
Tunnel0          172.16.0.2      YES manual up          up
Vlan1            unassigned      YES unset  administratively down down
Router#
```

PING in PC 0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=7ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=7ms TTL=126

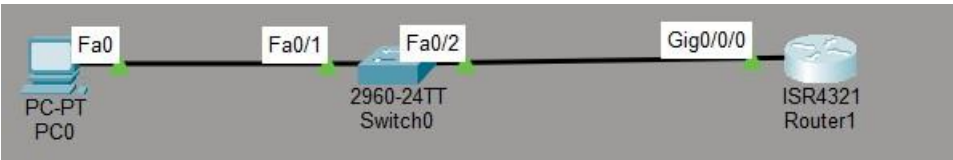
Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms

C:\>
```

The site-to-site GRE VPN was configured successfully, and branch-to-branch connectivity was verified using ping.

- 3. Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.
Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.

To simulate a **Remote Access VPN** scenario in Cisco Packet Tracer by configuring a VPN client PC to securely connect to a branch router using the built-in **VPN Client** feature.



IP Addressing

Device	Interface	IP Address	Subnet Mask
--------	-----------	------------	-------------

Router0	G0/0/0	192.168.10.1	255.255.255.0
PC0	FastEthernet0	192.168.10.10	255.255.255.0

Router Interface Configuration

enable configure terminal interface

GigabitEthernet 0/0/0 ip address

192.168.10.1 255.255.255.0 no shutdown

exit

VPN User Configuration

username vpnuser password cisco123 aaa new-model

aaa authentication login VPN_AUTH local aaa

authorization network VPN_GROUP local VPN

Group Configuration

crypto isakmp client configuration group VPNGROUP key VPNKEY

pool VPNPOOL

exit

ip local pool VPNPOOL 192.168.20.10 192.168.20.20

access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255

ISAKMP Configuration crypto isakmp policy 10

encryption aes hash sha

authentication pre-share group 2

exit

crypto isakmp key VPNKEY address 0.0.0.0

IPsec Configuration crypto ipsec transform-set

VPN_SET esp-aes esp-sha-hmac crypto dynamic-map

VPN_DYNAMIC 10 set transform-set VPN_SET

exit

crypto map VPN_MAP 10 ipsec-isakmp dynamic VPN_DYNAMIC interface

GigabitEthernet 0/0/0 crypto map VPN_MAP

exit

```
crypto map VPN_MAP client authentication list VPN_AUTH crypto map
VPN_MAP isakmp authorization list VPN_GROUP crypto map VPN_MAP
client configuration address respond
```

○ VPN Client Configuration On Pc0

VPN Parameter	Value
Group Name	VPNGROUP
Group Key	VPNKEY
Host IP (Server IP)	192.168.10.1
Username	vpnuser
Password	cisco123

○ Verification show crypto map

The basic network connectivity was successfully verified. The VPN connection itself requires a more realistic topology with the VPN client located on a **separate remote/WAN network** from the branch VPN server.

4. List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.



Feature	Site-to-Site VPN	Remote Access VPN
Connection	Router ↔ Router	PC/User ↔ VPN Router
Purpose	Connects two entire networks/branches	Allows an individual user to access a private network remotely
VPN endpoint	Usually two routers/firewalls	One VPN server/router and a VPN client
User authentication	Usually router-to-router authentication using pre-shared keys or certificates	Usually requires username/password plus VPN group/key
Client software	Normally no special software is required on individual PCs	A VPN client is required on the user's device
Example	Head Office ↔ Branch Office	Employee laptop → Branch Office
Configuration in Packet Tracer	Both routers are configured for IPsec	Router is configured as VPN server and PC uses the built-in VPN client

Site-to-Site VPN Advantages & Disadvantages Advantage:

Connects entire networks: All devices in both branches can communicate through the VPN.

Easy for users: Employees don't need to start a VPN connection individually because the router handles the tunnel.

Disadvantage:

More expensive infrastructure: Both locations need VPN-capable routers/firewalls.

Less suitable for individual remote users: It is designed for connecting fixed networks rather than employees working from different locations.

Remote Access VPN Advantages & Disadvantages

Advantages:

1. Supports remote workers: Users can access the company network from outside the office.
2. Individual authentication: Each user can have separate credentials, providing better control over who can access the network.

Disadvantages:

1. Requires VPN client configuration: Every remote device needs to be configured with the appropriate VPN settings.
2. Depends on the user's Internet connection: Poor or unstable Internet connectivity can affect the VPN connection and performance.

5. Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

To configure a Site-to-Site VPN in Cisco Packet Tracer so that two different branch networks can communicate securely through a VPN tunnel using GRE, IPsec, and routing.



2. Network Topology

PC1 — Switch1 — Router0 **=====** Router1 — Switch2 — PC2 VPN Tunnel

3. IP Addressing

Device	Interface	IP Address	Subnet Mask
Router0	G0/0	192.168.1.1	255.255.255.0
Router0	G0/1	10.0.0.1	255.255.255.252
Router1	G0/1	10.0.0.2	255.255.255.252
Router1	G0/0	192.168.2.1	255.255.255.0
Router0	Tunnel0	172.16.0.1	255.255.255.252
Router1	Tunnel0	172.16.0.2	255.255.255.252
PC1	Fa0	192.168.1.10	255.255.255.0
PC2	Fa0	192.168.2.10	255.255.255.0

4. Router0 – Interface Configuration

```
enable
configure terminal
interface gigabitEthernet 0/0 ip address 192.168.1.1 255.255.255.0
no shutdown exit
interface gigabitEthernet 0/1 ip address 10.0.0.1
255.255.255.252 no shutdown exit
```

5. Router1 – Interface Configuration

```
enable
configure terminal
interface gigabitEthernet 0/0 ip address 192.168.2.1 255.255.255.0
no shutdown exit
interface gigabitEthernet 0/1 ip address 10.0.0.2
255.255.255.252 no shutdown exit
```

6. GRE Tunnel – Router0

```
interface tunnel 0 ip address 172.16.0.1 255.255.255.252 tunnel
source gigabitEthernet 0/1 tunnel destination 10.0.0.2 no shutdown
exit
```

7. GRE Tunnel – Router1

```
interface tunnel 0 ip address 172.16.0.2 255.255.255.252 tunnel
source gigabitEthernet 0/1 tunnel destination 10.0.0.1 no shutdown
exit
```

8. Static Routing – Router0 ip route 192.168.2.0 255.255.255.0 172.16.0.2

9. Static Routing – Router1 ip route 192.168.1.0 255.255.255.0 172.16.0.1

10. IPsec Configuration – Router0

```
crypto isakmp policy 10 encryption aes hash sha authentication pre-share
group 2 exit
crypto isakmp key VPNKEY address 10.0.0.2
crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
```

```
crypto map VPN_MAP 10 ipsec-isakmp
set peer 10.0.0.2

set transform-set VPN_SET match address 100 exit interface
gigabitEthernet 0/1 crypto map VPN_MAP exit
```

11. IPsec Configuration – Router1

```
crypto isakmp policy 10 encryption aes hash sha
authentication pre-share group 2 exit crypto isakmp
key VPNKEY address 10.0.0.1

crypto ipsec transform-set VPN_SET esp-aes esp-sha-hmac
access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
crypto map VPN_MAP 10 ipsec-isakmp set peer 10.0.0.1

set transform-set VPN_SET match address 100 exit interface
gigabitEthernet 0/1 crypto map VPN_MAP exit
```

12. PC1 Configuration

IP Address: 192.168.1.10 Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

13. PC2 Configuration

IP Address: 192.168.2.10 Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

14. Verification Commands

```
show interfaces tunnel 0 show ip route show crypto
isakmp sa show crypto ipsec sa show crypto map
```

15. Connectivity Test

From PC1:
ping 192.168.2.10
From PC2:
ping 192.168.1.10

A Site-to-Site VPN was configured between Router0 and Router1 using a GRE tunnel protected by IPsec. Static routes were added so that the two branch LANs could communicate through the VPN. The configuration can be verified using tunnel, routing, ISAKMP, IPsec, and end-to-end ping commands.